

Reading Unseen Scripts: A Zero-Real-Image Evaluation Protocol and a Scored Forecasting Campaign for Brahmic Scene Text

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Abstract

Scene-text recognition is effectively solved for Latin script and fails for most of the world’s writing systems, yet no benchmark scores the case that decides the long tail: a script for which no real labeled image exists at all. We contribute a protocol for that case and the infrastructure to audit it, rather than a new dataset. It is leave-one-script-out over nine Brahmic scripts of a public benchmark: the held-out script appears only as rendered synthetic images, never a photograph, and is read in a shared grapheme pivot space whose inverse we audit rather than assume (100.000% exact recovery on single-script labels), so pivot accuracy is target-script accuracy. An out-of-benchmark rung extends it to Khmer, a disjoint Unicode block. We release the pivot construction, the rendering pipeline, every split and result, and a harness that re-derives all 169 result numbers here and fails on mismatch; a tenth script needs only a correspondence map, a word list and a font.

Applied to a compact open vision–language model, the protocol makes every held-out script readable at 9.16–30.09% word recognition, from source-only baselines of 0.0–5.46%, and places it against three reference systems on the identical crops under the identical metric: stock per-script Tesseract, trained on real images of each target script (14.35%); a prompted 7B VLM (7.53%); and the benchmark authors’ supervised specialists (66.26%), of which we recover 29%. Swapping the pivot for a stock BPE tokenizer under identical data, seed and recipe degrades all nine scripts (sign test $p = 0.0039$), isolating output-space structure as the mechanism.

Every forecast was committed before the run it predicts, and scoring them is what makes such an apparatus informative rather than decorative: it rules things out. The preregistered primary, that tokenizer fertility predicts transfer, is falsified ($\rho = -0.80$); a filed dose–

response is log-linear with a script-invariant slope but missed its committed magnitude by $5.1\times$; Khmer, forecast before we had trained any Khmer recognizer, returned 0.8%, a -3.66σ miss that locates the method’s boundary. We advance no predictive law. We advance the protocol that refuted these claims, the infrastructure to check it, and a scored record of what it ruled out.

1. Introduction

Modern scene-text recognizers exceed 90% word accuracy on Latin benchmarks, yet most of the world’s writing systems remain unreadable to them. Recent audits are stark: across more than a hundred Unicode scripts, even frontier vision–language models (VLMs) read fewer than thirty, and on unfamiliar scripts they emit garbled or substituted output [17]. For the Brahmic family, nine major scripts used by over a billion people, real-image training data exists only because of recent, labor-intensive benchmark efforts [9, 23], and dozens of related scripts (Sinhala, Khmer, Myanmar, Tibetan, ...) have little or none. Collecting and annotating real photographs script-by-script is not a scalable path to coverage, and, crucially, no existing benchmark measures how well a recognizer does when a script’s real labeled images are absent entirely. That gap is an evaluation gap before it is a modeling one.

This paper is about closing it. We ask a question existing protocols cannot score, can a recognizer read a script for which no real labeled image exists at all?, and build the machinery to answer it honestly: a protocol isolating the zero-real-image condition, a forecasting discipline that commits every prediction before its run, and a verification chain that lets a reader re-derive every number. The protocol is leave-one-script-out (LOSO) over all nine scripts of a public real-image benchmark [9]: for each held-out script we train a source-only model (Rung A) and a

075	source+synthetic-target model (Rung B), both evalu-	cast is auditable, hits and misses alike (Kannada	128
076	ated on the held-out script's real test images. A fur-	$\sim 15 \rightarrow 15.42$; Gurmukhi 16.2 filed 3.7 h pre-training	129
077	ther out-of-benchmark probe holds out Khmer, whose	$\rightarrow 22.16$; Khmer 16.1 $\rightarrow 0.8$, a -3.66σ miss).	130
078	Unicode block and coeng-based stacking sit outside the	3. Controlled attribution of the effect — the pivot is	131
079	nine. It returns 0.8% WRR against a 16.1% forecast	the mechanism. A preregistered stock-BPE abla-	132
080	filed before we had trained one, locating the proto-	tation under identical data, seed, and recipe degrades	133
081	col's boundary as sharply as its successes. The recog-	WRR on all nine scripts ($1.08 \times - 3.91 \times$; sign test	134
082	nizer exercising this machinery is a compact open	$p = 0.0039$), isolating output-space structure, not	135
083	VLM (Florence-2, 0.23B [29]) fine-tuned in a shared	synthetic fine-tuning per se, as the driver; a from-	136
084	grapheme pivot space: Brahmic scripts share an or-	scratch CRNN replicates the direction across an ar-	137
085	ganizational scheme (consonant bases, vowel signs,	chitecture with no large-scale pretraining but not	138
086	virama-formed conjuncts), so each script's text decom-	the magnitude, and we report the ambiguity rather	139
087	poses into grapheme clusters mapping by structural	than resolve it in our favor.	140
088	correspondence into one output vocabulary. The pivot	4. System analysis on the benchmark, and released re-	141
089	is the object the evaluation measures, not the headline	sources. On the identical test crops we situate a	142
090	claim.	prompted 7B frontier VLM (reads two of nine; loses	143
091	Our contribution class is an evaluation protocol plus	to the compact model on eight of nine), off-the-	144
092	verification infrastructure, not a data release: we col-	shelf per-script OCR (14.35%, which our zero-real-	145
093	lect no new photographs, instead repurposing two pub-	image model beats on eight of nine), and the bench-	146
094	lic benchmarks [9, 25] under a new held-out condi-	mark authors' own supervised specialists (66.26%,	147
095	tion and adding verified synthetic renderings. This is	of which we recover 29%), giving an interpretable	148
096	squarely what this track solicits: new evaluation proto-	floor-to-ceiling context, alongside the released pro-	149
097	cols, rigorous auditing and stress-testing, and negative	protocol, harness and preregistration chain (Sec. 3.3).	150
098	results, all of which it invites independently of new	Auditing the benchmark's labels in passing turns up	151
099	datasets.	30 test items that mix two Brahmic blocks; all three	152
100	Our contributions, in the order this paper weights	systems above score 0 on them.	153
101	them:		
102	1. A zero-real-image evaluation protocol. A LOSO	2. Related Work	154
103	scene-text protocol that measures reading an entire	Multilingual STR and Indic benchmarks. STR is ma-	155
104	writing system, disjoint Unicode block and disjoint	ture for Latin script [4] but collapses elsewhere. Indic-	156
105	glyph inventory, from zero real target images, in a	STR12 [23] and the Bharat Scene Text Dataset [9] es-	157
106	shared pivot space whose inverse mapping we au-	tablished real-image benchmarks for Indian languages;	158
107	dit rather than assume (100.000% exact recovery	on the latter, PARSeq reaches $\sim 47\%$ average WRR	159
108	on single-script labels, so pivot-WRR equals target-	from synthetic data alone and $\sim 73\%$ fine-tuned on real	160
109	script WRR), plus an out-of-benchmark generaliza-	labeled images, and in-the-wild collections keep grow-	161
110	tion probe (Khmer). It surfaces a strong effect exist-	ing the real-image supply one language at a time [28].	162
111	ing protocols never test: every held-out script moves	All of it presupposes labeled target data. Transfer be-	163
112	from a $\leq 5.46\%$ source-only baseline to 9.16–30.09%	tween Indian languages is established practice: Gunna	164
113	WRR with zero real target images.	et al. [13] show that transferring among Indic scripts	165
114	2. Prospective prediction as an evaluation practice —	beats transferring from English, exploiting the shared	166
115	and a mostly negative record. A frozen, version-	vowels and conjuncts our pivot formalizes, and follow-	167
116	controlled preregistration with committed-before-	up work adds font diversity to that recipe [12]. Both	168
117	run forecasts, and a discipline of scoring every call	fine-tune on real images of the target language — the	169
118	including its misses. We advance no predictive	target is never absent. GlotOCR [17] measures the	170
119	law — the record is the contribution, and it is	wider problem at scale, finding that frontier VLMs	171
120	largely one of failures. The preregistered primary	read fewer than thirty of 100+ scripts, but it scores	172
121	(tokenizer fertility predicts transfer) was falsified	pretrained models on rendered text. That leaves the	173
122	(Spearman -0.80); a prospectively filed synthetic-	training-side question open: not what a model already	174
123	exposure study gives a script-invariant, log-linear	knows, but what it can be taught about a script of	175
124	dose-response ($+2.235$ WRR/doubling, F -test $p =$	which no real image exists. Synthetic-data engines re-	176
125	0.45) whose pre-committed magnitude missed by	main Latin-centric [30], and synthetic-rendering fine-	177
126	$5.1 \times$; pivot coverage orders scripts in-sample but	tuning has been shown for a single low-resource script	178
127	not out-of-sample ($r = +0.03$). Every point fore-	with real-data validation [7]. We target the regime	179

where no real labeled target image exists at all.

Tokenization granularity for complex scripts. MGP-STR [8] fuses character, BPE and WordPiece streams in one Latin recognizer; grapheme tokenization beats byte-BPE for Bengali handwriting [14]; and fertility’s value as a tokenizer metric is contested in the LLM literature [24]. A companion manuscript by the same authors [1], anonymized in the supplement, makes the supervised grapheme-vs-BPE choice predictable across nine scripts (polarity 8/9, leave-one-script-out 90.9%); the present paper shares no experiment with it, and tests, then refutes, the conjecture that the same property predicts zero-real-image transfer.

Unseen units vs. unseen scripts. A substantial line reads unseen characters within a known script by decomposition into shared sub-units: stroke- and radical-based zero-shot Chinese recognition [5, 31, 32] and open-set recognition of novel characters [20–22]. Zero-shot detection of unseen scripts localizes but does not read [18]. Byte-level pivots let ASR cover unseen-script languages [6], and task-arithmetic analogies achieve zero-real-data synthetic-to-real HTR for five Latin-script languages, leaving unseen writing systems explicitly open [11]. To our knowledge no prior system reads real scene text of an entirely unseen script — different Unicode block, different glyphs — from zero real target images.

What predicts cross-lingual transfer. In NLP, transfer tracks lexical and subword overlap rather than language relatedness [10, 26]. For STR specifically, Ref. [2] concludes that source dataset size and visual similarity, not linguistic typology, drive cross-lingual transfer. Our findings sharpen rather than contradict this, in a regime that work did not study (zero real target images): (i) a typology-adjacent structural property, fertility, indeed fails as a transfer predictor ($\rho = -0.80$), as a data-centric view expects; (ii) synthetic target exposure dominates, which is a target-side data-size effect; (iii) within a fixed budget, coverage of the shared grapheme space orders transfer ($\rho = +0.61$ among equal-budget scripts), an OCR instantiation of the subword-overlap principle in a designed output space; and (iv) a frozen-encoder visual-similarity control predicts nothing in our data ($\rho = -0.12$). Because LOSO balances source data by construction, the confound identified by [2] is controlled by design.

Prediction and preregistration in empirical ML. Scaling laws make supervised STR predictable in model and data size [27]; preregistration and negative-result

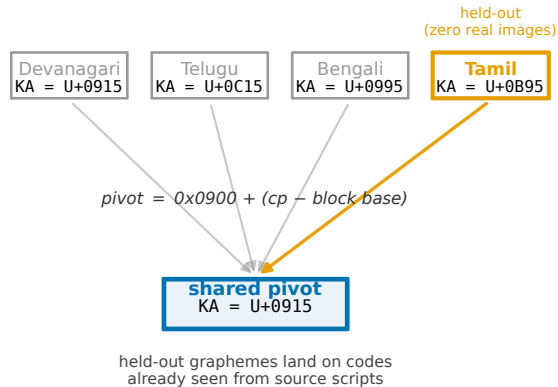


Figure 1. The shared grapheme pivot space. Each Brahmic script maps bijectively into a common output space by ISCII-derived Unicode-offset correspondence (the same letter sits at the same offset within each block); grapheme clusters in that space form the recognizer’s output vocabulary. A held-out script’s graphemes therefore land on pivot codes already seen from the source scripts. The held-out script enters training only as verified synthetic renderings (Rung B) or not at all (Rung A); evaluation is always on its real scene images.

reporting are the reproducibility literature’s recommended remedies for HARKing and cherry-picking [15, 16], yet full-study adoption is rare. We run the entire campaign under such a preregistration and score every forecast it filed.

3. Method

3.1. A shared grapheme pivot space for Brahmic scripts

Brahmic scripts encode the same phonological skeleton (consonant bases, dependent vowel signs, and conjunct formation through a virama), and Unicode preserves this: corresponding letters sit at the same offset within each script’s block. We exploit both layers.

Structural pivot. A mapping π sends any Brahmic codepoint to a pivot codepoint by its within-block offset (Fig. 1; $\pi(c) = 0x0900 + (c - \text{base}(c))$); anything outside the Brahmic blocks (Latin, ASCII digits, punctuation, zero-width joiners) passes through untouched. Two script-neutral marks encoded inside the Devanagari block, the danda and double danda, also pass through unshifted, since their offsets are unassigned in every other Brahmic block and cannot collide. The OM and abbreviation signs deliberately do not get that treatment: their offsets carry real letters elsewhere (Assamese RA, Gurmukhi TIPPI), so they are shifted like

any other letter. Because π is a per-script bijective shift, it is exactly invertible: a hypothesis decoded in pivot space is read back into the target script deterministically. Text from nine scripts thereby writes into one label space in which structurally corresponding syllables become identical strings.

The round trip, measured. Invertibility is a property of the construction, so we checked it against the data rather than asserting it. Mapping all 16178 scored test labels into pivot space and back recovers 99.955% of characters and 99.815% of words; on labels written wholly in the script they are filed under, recovery is 100.000% for all nine. The whole residual is 30 labels (0.19%, 17 of them Gurmukhi) mixing two Brahmic blocks: 7 words typed in the wrong script, and 23 where a homoglyph was substituted inside a correct one, such as a Devanagari digit for its Gujarati twin. These are benchmark annotation faults, invisible in the rendered image, and they are unwinnable under exact match rather than merely hard: our Rung B, the Tesseract floor and the supervised PARSeq ceiling each score 0 of 30 on them. We leave them in and count them as errors, keeping our numbers conservative and comparable with published ones.

Grapheme-cluster vocabulary. Within pivot space we segment text into grapheme clusters (consonant-virama conjuncts kept whole, dependent signs attached) and inject the resulting cluster inventory (1624 types for the source pool) as dedicated tokens into the recognizer’s tokenizer. The alternative, letting the stock byte-BPE tokenizer segment pivot-space text, is the ablation of Sec. 4: it sees the same pivot strings but fragments clusters into sub-syllabic pieces. Fertility (mean clusters per word) governs when grapheme tokenization beats BPE in supervised training: in a 12-run balanced-budget study across the same nine scripts, fertility predicts the winning branch with 8/9 polarity and 90.9% leave-one-script-out direction accuracy ($R^2=0.678$). That result motivates both the pivot design and the (ultimately falsified) primary hypothesis of Sec. 5.

3.2. Zero-real-image protocol

For each held-out script S of the nine in the benchmark [9]:

Sources. All benchmark languages whose script is not S (languages sharing S , e.g. Assamese for Bengali script, are excluded, so S is genuinely unseen). Each source language contributes exactly 700 train / 90 val real images (seed 42), giving balanced source exposure

by construction. Source-side data size is therefore controlled, unlike prior cross-lingual STR studies [2].

Synthetic target. 3240 images per target language: benchmark-vocabulary words taken from that language’s train split only, never the test split, rendered in fontconfig-verified fonts (multiple faces per script, libraqm shaping), each render checked non-blank, and the rendered word enters training as its pivot transcription. No real photograph of S is ever seen. Scripts written by two source languages receive two languages’ worth of renderings ($2\times$ budget), a covariate disclosed at preregistration time that becomes central in Sec. 5. Rungs. Rung A trains on sources only. Rung B adds the synthetic target set, and Rung B_{bpe} repeats Rung B with the grapheme-cluster injection removed (stock tokenizer), identical data, seed, and recipe. All rungs fine-tune Florence-2 (0.23B) [29] with LoRA ($r=16$, $\alpha=32$, dropout 0.05), lr 10^{-4} , batch size 4, 15 epochs, validation-selected checkpoint, beam-search decoding at test.

Evaluation. The held-out script’s real test split, WRR = exact match after NFC normalization, zero-width-character removal, and whitespace collapse, which is the benchmark’s own protocol [9, 23], plus character accuracy ($1 - \text{CER}$).

3.3. Preregistration and prospective predictions

The campaign ran under a frozen preregistration with an append-only amendment log — every amendment was filed before the results it concerns existed. Descriptors entering the transfer horse-race were computed result-blind: fertility and pivot coverage before any Rung-B result, and a frozen-encoder visual-similarity control before the third. Quantitative forecasts for later runs were committed and pushed to a version-controlled archive before those runs trained, and are scored against a rule fixed at filing time, hits and misses alike (Sec. 5). We state the limit of this plainly, because it is the weakest link in the design. The archive is a version-controlled repository that was private during the campaign: no third party observed the commits as they were made, and git’s own timestamps are settable by the author. The supplement ships the filed documents under SHA-256, which establishes that they are unaltered — not when they were written. Filing order is therefore attested rather than proven, and we do not present it as proven.

What survives that limit is worth separating from what does not. Every scoring rule, band and point value is in the supplement and checkable against the results, so a reader can confirm each forecast is scored by the rule filed with it rather than one chosen later. The shape of the record is itself evidence: documents

written after the fact would not have produced a falsified primary hypothesis, a $5.1\times$ magnitude miss and a -3.66σ out-of-block miss. Backdating buys hits, not this. The forecasts are genuine. Their timing is unwitting.

The general lesson is a protocol one, and it may be the more useful contribution. A campaign can be run honestly and still end up unverifiable, which is the exact failure this apparatus exists to prevent. Anchor each filing, as it is filed, in a timestamp no author controls: an anonymized public preregistration, or a hash on an append-only ledger. Ours predates that decision; anyone building on this protocol should anchor their filings from the start.

Released resources. We release the apparatus, not a description of it: the pivot construction and its 1624-type grapheme vocabulary; the verified rendering pipeline; every LOSO split actually used (nine scripts $\times$ rungs A/B/ B_{bpe} , plus the Khmer rungs); every per-run result file the paper cites, anchors included; the preregistration chain with each forecast and its scoring receipt; and `verify_wacv_numbers.py`, which rederives every derivable number in this paper from those result files and exits non-zero on any mismatch (169 macros, zero mismatches at submission). Extending the protocol to a tenth script needs only a correspondence map, a word list and a font; the Khmer rung is the worked example, and the harness is unchanged.

4. Experiments

Our comparisons are the source-only baseline (Rung A), the raw pre-trained VLM (WRR 0.0), the stock-BPE ablation, the reference systems of Sec. 4.2, and our own forecasts.

4.1. Main result: a strong effect existing protocols never test

Every one of the nine moves from a near-floor source-only baseline (0.0–5.46% WRR; the raw pre-trained model reads none of them) to 9.16–30.09% WRR at 34.57–57.77% character accuracy, without a single real target image. Transfer holds even where there is least to borrow: Malayalam has the lowest pivot coverage (79.2%) and still goes 0.18 $\rightarrow$ 9.69. Rung A tracks incidental pivot-space coverage, highest for Devanagari, so the pivot shares structure before any target exposure.

4.2. Floor and ceiling

Absolute numbers need a scale, and the supervised ceiling alone is the wrong one. Against the ceiling we recover 0.29 of the macro-average (19.41% vs. 66.26%):

Script	<i>N</i>	A	B	B 95% CI	CharAcc	B_{bpe}
Tamil	513	0.0	9.16	[6.6, 11.7]	39.0	2.34
Telugu	545	1.28	19.82	[16.5, 23.1]	54.43	10.28
Kannada	720	2.78	15.42	[12.8, 18.1]	46.37	13.33
Malayalam	547	0.18	9.69	[7.3, 12.2]	34.57	4.2
Oriya	1044	0.77	25.38	[22.7, 28.1]	56.37	14.94
Gujarati	1015	3.05	15.86	[13.6, 18.2]	38.49	13.69
Bengali	2873	1.39	27.08	[25.5, 28.7]	57.77	19.07
Devanagari	6042	5.46	30.09	[28.9, 31.3]	57.16	27.99
Gurmukhi	2879	0.59	22.16	[20.7, 23.7]	44.1	17.65
macro-average			19.41			

Table 1. Zero-real-image transfer across nine held-out Brahmic scripts (real test images). Rung A: source scripts only. Rung B: + synthetic target renderings, zero real target images (95% CI = percentile bootstrap over test samples, 5000 resamples). B_{bpe} : Rung B with the grapheme-cluster vocabulary removed (stock tokenizer), identical data and seed. On 6/9 scripts the Rung-B CI lies strictly above the B_{bpe} 95% CI. Reference systems on these same crops are in Table 2.

Script	Tess.	Qwen	Rung B	PARSeq	o/c
Tamil	15.4	6.24	9.16	76.61	0.12
Telugu	13.76	0.18	19.82	55.78	0.36
Kannada	12.08	0.28	15.42	59.44	0.26
Malayalam	5.85	0.0	9.69	64.53	0.15
Oriya	15.52	0.38	25.38	67.62	0.38
Gujarati	12.41	0.3	15.86	57.04	0.28
Bengali	15.32	24.82	27.08	72.82	0.37
Devanagari	19.78	35.05	30.09	68.9	0.44
Gurmukhi	19.03	0.52	22.16	73.6	0.30
macro-avg	14.35	7.53	19.41	66.26	0.29

Table 2. Every reference system on the identical crops, through the identical pivot mapping and exact-match metric. Tess. — stock per-script Tesseract 5, trained on real labeled images of each target script (off-the-shelf floor). Qwen — Qwen2.5-VL-7B prompted zero-shot, scored under the disclosed most-favorable extraction. PARSeq — the benchmark authors’ own supervised specialists, also trained on real labeled target images; BSTD train and test share no source image on any script, and the measured mean is in line with the $\sim 73\%$ they report. o/c = ours / ceiling. Both trained-on-real systems presuppose exactly the data whose absence defines our setting.

real labels remain worth roughly $3.4\times$ what our synthetic pivot buys, and we do not present that as close. Against the floor the picture inverts. Stock Tesseract ships a per-script model for every script here, trained on real target images, and is what a practitioner deploys today: it reaches 14.35%, and our zero-real-image model is ahead by +5.06 points on macro-average, los-

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ing only Tamil. It is a document engine outside its design domain, so this is no claim to be the better recognizer, only evidence that the zero-real-image regime has passed what is obtainable off the shelf.

4.3. The pivot is the mechanism: BPE ablation

Removing the grapheme-cluster vocabulary, with the same pivot strings, images and seed, degrades all nine scripts (Table 1), from $1.08\times$ (Devanagari, 30.09 $\rightarrow$ 27.99 WRR) to $3.91\times$ (Tamil, 9.16 $\rightarrow$ 2.34). The margin varies, the direction never does (exact two-sided sign test, $p = 0.0039$). The bootstrap makes the per-script margin concrete: on 6/9 scripts the Rung-B 95% CI clears the B_{bpe} interval outright (Table 1); the three exceptions, Devanagari, Kannada and Gujarati, are exactly the near-parity scripts ($1.08\text{--}1.16\times$) — consistent with the direction holding but the magnitude shrinking as fertility falls. The character-accuracy pattern is diagnostic: on Telugu, Kannada and Devanagari the BPE model equals or exceeds the pivot model’s character accuracy (Telugu 55.1% vs. 54.43%), yet word accuracy drops in every case. Synthetic fine-tuning alone teaches the model to see the characters. The cluster-level output space is what assembles them into correct whole words.

4.4. Dose-response: synthetic budget

We hold the recipe fixed and vary only the number of synthetic target images, for the three scripts whose held-out budget we can grow (Malayalam, Kannada, Telugu) across {810, 1620, 6480, 12960} renderings; the external Rung-B point (3240) anchors each curve. Word recognition is cleanly log-linear in synthetic exposure (R^2 from 0.905 to 0.999 per script), and the slope is script-invariant: a single shared $+2.235$ WRR per doubling fits all three, with no evidence of differing slopes (F -test $p = 0.45$; Fig. 2). This is the preregistered form, and it holds.

The preregistered magnitude does not. The frozen instrument projected $+11.48$ WRR per doubling, transported from the in-script $2\times$ -synthetic experiment that fit the two-factor model (Sec. 5). The swept slope excludes that by a factor of 5.1, with the faithful doubling step ($3240 \rightarrow 6480$) gaining $+2.35$ WRR against a predicted $+11.5$, outside the ± 2 RMSE band and reported as filed (open squares, Fig. 2). Two further preregistered edges resolve the same way: the low-budget prediction of near-collapse toward the Rung-A floor is falsified (Kannada and Telugu already reach double digits at 810 images, so transfer is concave in log-budget), and the top step ($6480 \rightarrow 12960$), which adds renderings at near-fixed lexicon, gains little on Kannada but keeps pace on Malayalam and Telugu. Syn-

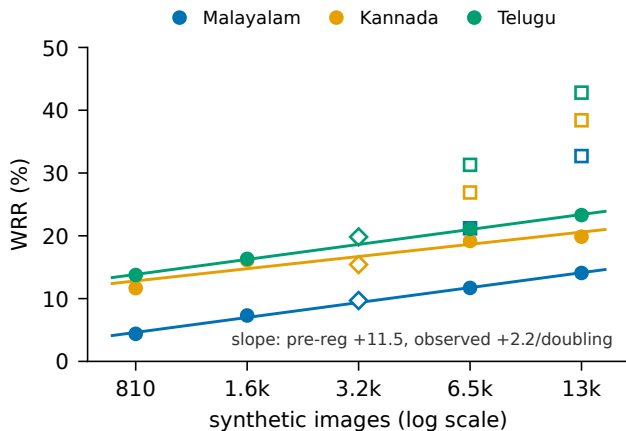


Figure 2. Synthetic-budget dose-response for three held-out scripts. Filled circles: observed WRR on real test images; solid lines: per-script log-linear fits (shared slope $+2.235/\text{doubling}$); open diamonds: the external Rung-B anchor (3240 images); open squares: the preregistered point predictions — their gap above the observed points is the $5.1\times$ magnitude miss, reported as filed.

thetic scaling is real and orderly, then, but far shallower than an in-script extrapolation predicts. We do not promote pivot-space coverage to a cross-script driver: it orders the three swept scripts in-sample, but across the six never used in the fit it is uncorrelated with transfer ($r = +0.03$).

4.5. Architecture generality: a preregistered CRNN replication

Every result above uses one backbone, inviting the objection that the effect is specific to a pretrained vision-language model. We preregistered a replication (Amendment 6, committed before any CRNN trained) on a from-scratch CRNN (ResNet+BiLSTM, CTC, no large-scale pretraining) over the same LOSO splits, pivot space and metric, for the three equal-synthetic scripts spanning the observed Rung-B range (Tamil, Telugu, Oriya), vocabulary from TRAIN records only and a 60-epoch budget fixed in advance. Four predictions were filed; all four are scored.

The two structural predictions hold: source-only Rung A stays at the floor on all three ($\text{WRR} \leq 0.67$, predicted < 2.0), and the held-out ranking is preserved ($0.97 < 1.1 < 4.5$, matching the Florence-2 order), with Rung B above Rung A everywhere. The two effect-size predictions miss. The preregistered primary, a $\geq +2.0$ WRR lift on at least two of three, holds only on Oriya ($+3.83$). The CRNN reaches a $0.11\times$ fraction of the Florence-2 transfer level on average (predicted band $[0.15, 0.60]$), inside the band on Oriya alone, and character accuracy is depressed in step (Tamil 25.09%

491 vs. 39.0%): the small model sees fewer characters, not
492 merely fewer whole words. Per the preregistration’s de-
493 clared caveat we call this ambiguous rather than resolve
494 it in our favour: an attenuated directional replication
495 fits both a backbone-dependent mechanism and a plain
496 capacity floor, and the character accuracy supports the
497 latter. The direction replicates across architectures;
498 the magnitude stays open.

499 4.6. Out-of-benchmark: Khmer

500 Every script above shares the ISCII-aligned Brahmic
501 design the pivot was built around, so to find where
502 the method stops we ran it somewhere it should strug-
503 gle: Khmer, a disjoint Unicode block with coeng-based
504 stacking, outside the benchmark and outside India.
505 Reaching it needed a hand-curated pivot extension
506 (94.41% of test codepoints map; the rest are approx-
507 imated or dropped and counted as misses). Under
508 Amendment 4b, Rung A trains on the nine benchmark
509 scripts with no Khmer of any kind and Rung B adds
510 3240 synthetic Khmer words, still with no real Khmer
511 image; both are evaluated on 1252 KhmerST [25] word
512 crops. Khmer recognition is an active area, but its sys-
513 tems train on real Khmer [19], exactly the data this
514 rung withholds. Before any Khmer recognizer of ours
515 existed we filed a point prediction from the frozen two-
516 factor model, 16.1 WRR at ± 1 -RMSE [11.9, 20.3], with
517 a written caveat that we expected it to be fragile.

518 It was. Rung B reaches 0.8% WRR against a pre-
519 dicted 16.1: a residual of -15.3, or -3.66-RMSE, below
520 even the ± 2 -RMSE band [7.8, 24.5]. The miss is not an
521 artifact of the coverage recheck disclosed at build time:
522 re-evaluating the same instrument at the realized token
523 coverage (88.01% rather than the filed 89.02%) moves
524 the point only to 15.52. Only the structural prediction
525 survives: Rung A sits at 0.0 (predicted < 5). The in-
526 strument had already lost its slope magnitude in block
527 (Sec. 4.4); this is the first script we asked it to predict
528 outside the blocks it was fit on, and it failed there too.
529 We do not carry it forward as a contribution.

530 Two things nonetheless hold. The direction repli-
531 cates (+0.8 WRR, +5.85 character accuracy over the
532 no-target-data control), so the mechanism is not in-
533 ert out of block; the magnitude collapses to 4.1%
534 of the nine-script Rung-B mean. For calibration,
535 stock Tesseract’s Khmer-specific model, trained on real
536 Khmer and scored on the identical crops, reaches only
537 2.16% here against 14.35% across the nine: these crops
538 are hard for a supervised system too. That is context,
539 not a rescue. Both numbers sit near the floor, and 0.8%
540 is a failure to transfer usefully.

541 We draw the boundary claim narrowly. Khmer re-
542 ceived the same $1\times$ synthetic budget and comparable

nominal token coverage (88.01%) that yield double-
digit WRR in block, and still returned 0.8. Volume
and coverage were held up; typological proximity was
not; the transfer collapsed. That is the cleanest evi-
dence we have that relatedness, not synthetic volume,
carries the effect, but it is a single script, and we state
it as suggestive rather than decisive.

4.7. Frontier-VLM reference

We prompt Qwen2.5-VL-7B [3] zero-shot on the same
nine real test sets (fixed prompt, greedy decoding), pre-
registered under Amendment 4c. Under the declared
primary metric, exact word match on the raw genera-
tion, it scores 0.0% on all nine, because the chat model
wraps every answer in prose (“The text in the image
is ...”). We therefore report a disclosed, most-favorable
re-score (Table 2): the first quoted span from each gen-
eration, under the identical metric. Even so the VLM
reads only the two highest-resource scripts with any
competence (Bengali 24.82%, Devanagari 35.05%), sits
at or near zero on six of the remaining seven, and of-
ten returns a wholly different script: Gurmukhi in De-
vanagari, Malayalam in Bengali. Our compact pivot
model leads on eight of nine at a fraction of the pa-
rameters. This is one open model under one prompt
and the comparison is not like-for-like, its training data
being unknown; the broader audit of [17], over 100+
scripts and many models, is the load-bearing evidence
that scale alone does not read these scripts.

5. What Predicts Transfer?

We report the confirmatory analysis exactly as prereg-
istered, our own primary hypothesis included.

5.1. The preregistered primary fails

Fertility, the property that governs the supervised
grapheme-vs-BPE trade-off (Sec. 3.1), was preregis-
tered as the primary transfer predictor (H3). It is re-
futed: Spearman $\rho = -0.80$ across the nine Rung-B
results, directionally opposite to the hypothesis. The
flagship case was called in advance by the declared com-
petitor: our prospective prediction file ranked Malay-
alam best under fertility (P1) and worst under coverage
(P2); it realized bottom-tier (9.69 WRR). We report
this as a genuine falsification, consistent with fertil-
ity skepticism in the LLM-tokenizer literature [24] and
with the data-centric view of STR transfer [2].

Post-hoc, fertility instead tracks who needs the
pivot: BPE comes closest to parity on the lowest-
fertility script and collapses on the highest ($1.08\times$ De-
vanagari, $3.91\times$ Tamil). That is unregistered, $n = 9$,
and used for nothing here.

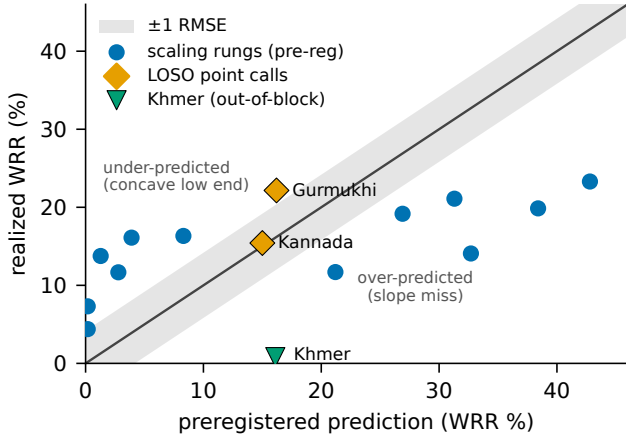


Figure 3. Prospective scorecard: preregistered prediction vs. realized WRR for every quantitative forecast filed before its run, scored against the rule fixed at filing time (shaded ± 1 RMSE). The in-block LOSO point calls land on the diagonal; the scaling runs fall off it both ways — under-predicted at low budget, over-predicted at high. Khmer, the out-of-block call, misses low by $-3.66 \cdot \text{RMSE}$. Hits and misses are both shown.

5.2. The two-factor account

Two result-blind quantities describe the campaign and generated its forecasts; we state them as description, not inference, because at $n = 9$ scripts nothing here carries appreciable inferential force. Synthetic budget dominates: the two scripts with the disclosed $2\times$ budget (Bengali, Devanagari) are the top two results (27.08, 30.09 WRR), Bengali from the second-lowest coverage. Coverage orders the seven equal-budget scripts ($\rho = +0.61$). The joint fit $\text{WRR} = -35.79 + 0.583 \cdot \text{cov} + 11.48 \cdot \mathcal{K}[2\times]$ (RMSE 4.18 WRR) is the instrument every later forecast was drawn from, so we print it for audit, and then discount it ourselves: a script-level bootstrap puts its coverage slope at $[-0.48, 3.49]$, spanning zero, and that term fails out-of-sample outright ($r = +0.03$, Sec. 4.4). Coverage is an in-sample ordering and nothing more. One control stands on its own: rendered glyphs embedded by the recognizer’s own frozen vision encoder predict transfer not at all ($\rho = -0.12$), separating whatever the ordering reflects from the visual-similarity account of [2].

5.3. Prospective scorecard

Four forecasts were filed before their runs and all four are scored in Fig. 3. Two are in-block point calls: Kannada (~ 15 WRR, filed with two scripts observed, realized 15.42) and Gurmukhi, the final script (16.2 WRR with $\pm 1\sigma$ [12.2, 20.2], committed and pushed 3.7 hours before training began, realized 22.16, under-called by

1.5σ but inside $\pm 2\sigma$). A third, Devanagari, was a directional call committed before the result existed but pushed after; it is only weakly prospective and we lean on it for nothing. The fourth is Khmer (Sec. 4.6), the one call filed outside the pivot’s Unicode blocks, which missed by -3.66σ . The declared rank bet between the two preregistered predictors resolved for coverage over fertility, neither ranking alone significant at $n = 7$.

The summary is bounded: in-block, transfer is forecastable to roughly ± 4.18 WRR (1σ) from two pre-training quantities, but the instrument missed out of block and we decline to extend it past the blocks it was fit on: one script shows it failed there, not that it must fail everywhere outside.

6. Limitations

Absolute accuracy. Our macro-average (19.41%) clears the off-the-shelf OCR floor (14.35%), though the weakest script does not: Tamil (9.16%) falls below it, and all nine sit far under the supervised ceiling ($\sim 73\%$ with real labels [9]). The synthetic $\rightarrow$ real domain gap is the dominant residual: training loss reaches ~ 0.06 while real-image WRR stays modest. The target script must be known. π^{-1} resolves a pivot string into one script, so the protocol assumes script identity is fixed before decoding. That is given for a held-out-script benchmark, and an extra component for mixed-script scenes. One family, and a located boundary. Nine scripts, one benchmark, one family whose shared structure the pivot exploits by design; Khmer (Sec. 4.6) shows the method failing outside it (0.8% WRR). One architecture. All transfer results use Florence-2; a from-scratch CRNN replicates the direction at ~ 0.11 of the magnitude (Sec. 4.5), so backbone-independence of the effect size is untested. Model imperfections. Gujarati underperforms its coverage, one Rung-A baseline breaches the predicted < 5 line (Devanagari, 5.46), and the final-script forecast under-called by 1.5σ ; all reported, none explained away. Compute. One 24 GB GPU; budgets were fixed by preregistration, not tuned.

7. Conclusion

We built an evaluation protocol for a condition existing benchmarks do not score, a script with no real labeled image at all, and ran a forecasting campaign inside it. Under it, a small open VLM fine-tuned in a shared grapheme pivot space on rendered synthetic text alone reads real photographs of all nine unseen Brahmic scripts at 9.16–30.09% word accuracy, and a stock-BPE ablation collapses that transfer, isolating output-space structure as the mechanism. The forecasting record is mostly one of misses: primary hypothesis falsified, scaling magnitude missed by

5.1 $\times$, coverage failing out-of-sample, the one out-of-block call (Khmer) missed by -3.66σ . We advance no predictive law. Every call ships in the supplement with its receipt, and the failures locate the method's boundary rather than leave it for someone else to find.

References

- [1] Authors. Predictable grapheme-vs-BPE tokenization for Brahmic script recognition. Parallel submission under review. Anonymized copy supplied as supplemental material parallel_submission.pdf, 2026. 3
- [2] Jeonghun Baek, Yusuke Matsui, and Kiyoharu Aizawa. Cross-lingual learning in multilingual scene text recognition. arXiv preprint arXiv:2312.10806, 2023. 3, 4, 7, 8
- [3] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report. arXiv preprint arXiv:2502.13923, 2025. 7
- [4] Darwin Bautista and Rowel Atienza. Scene text recognition with permuted autoregressive sequence models. arXiv preprint arXiv:2207.06966, 2022. 2
- [5] Jingye Chen, Bin Li, and Xiangyang Xue. Zero-shot chinese character recognition with stroke-level decomposition. arXiv preprint arXiv:2106.11613, 2021. 3
- [6] Zhehuai Chen, Ankur Bapna, Andrew Rosenberg, Yu Zhang, Bhuvana Ramabhadran, Pedro Moreno, and Nanxin Chen. Maestro-u: Leveraging joint speech-text representation learning for zero supervised speech asr. arXiv preprint arXiv:2210.10027, 2022. 3
- [7] Yan Hon Michael Chung and Donghyeok Choi. Fine-tuning vision-language models as ocr systems for low-resource languages: A case study of manchu. arXiv preprint arXiv:2507.06761, 2025. 2
- [8] Cheng Da, Peng Wang, and Cong Yao. Multi-granularity prediction with learnable fusion for scene text recognition. arXiv preprint arXiv:2307.13244, 2023. 3
- [9] Anik De, Abhirama Subramanyam Penamakuri, Rajeev Yadav, Aditya Rathore, Harshiv Shah, Devesh Sharma, Sagar Agarwal, Pravin Kumar, and Anand Mishra. Bharat scene text: A novel comprehensive dataset and benchmark for indian language scene text understanding. arXiv preprint arXiv:2511.23071, 2025. 1, 2, 4, 8
- [10] C. M. Downey, Terra Blevins, Dhvani Serai, Dwija Parikh, and Shane Steinert-Threlkeld. Targeted multilingual adaptation for low-resource language families. arXiv preprint arXiv:2405.12413, 2024. 3
- [11] Carlos Garrido-Munoz, Aniello Panariello, Silvia Cascianelli, Angelo Porrello, Simone Calderara, Jorge Calvo-Zaragoza, and Rita Cucchiara. Zero-shot synthetic-to-real handwritten text recognition via task analogies. arXiv preprint arXiv:2604.09713, 2026. 3
- [12] Sanjana Gunna, Rohit Saluja, and C. V. Jawahar. Improving scene text recognition for indian languages with transfer learning and font diversity. Journal of Imaging, 8(4):86, 2022. DOI 10.3390/jimaging8040086. 2
- [13] Sanjana Gunna, Rohit Saluja, and C. V. Jawahar. Transfer learning for scene text recognition in indian languages. arXiv preprint arXiv:2201.03180, 2022. 2
- [14] Md. Mahmudul Hasan, Ahmed Nesar Tahsin Choudhury, Mahmudul Hasan, and Md. Mosaddek Khan. Gradet-htr: A resource-efficient bengali handwritten text recognition system utilizing grapheme-based tokenizer and decoder-only transformer. arXiv preprint arXiv:2509.18081, 2025. 3
- [15] Moritz Herrmann, F. Julian D. Lange, Katharina Eggensperger, Giuseppe Casalicchio, Marcel Wever, Matthias Feurer, David Rügamer, Eyke Hüllermeier, Anne-Laure Boulesteix, and Bernd Bischl. Position: Why we must rethink empirical research in machine learning. arXiv preprint arXiv:2405.02200, 2024. 3
- [16] Jake M. Hofman, Angelos Chatzimpampas, Amit Sharma, Duncan J. Watts, and Jessica Hullman. Pre-registration for predictive modeling. arXiv preprint arXiv:2311.18807, 2023. 3
- [17] Amir Hossein Kargaran, Nafiseh Nikeghbal, Jana Diesner, François Yvon, and Hinrich Schütze. Glotocr bench: Ocr models still struggle beyond a handful of unicode scripts. arXiv preprint arXiv:2604.12978, 2026. 1, 2, 7
- [18] Prateek Keserwani, Taveena Lotey, Rohit Keshari, and Partha Pratim Roy. Separate scene text detector for unseen scripts is not all you need. arXiv preprint arXiv:2307.15991, 2023. 3
- [19] Marry Kong, Rina Buoy, Sovisal Chenda, Nguonly Taing, Masakazu Iwamura, and Koichi Kise. Towards universal khmer text recognition. arXiv preprint arXiv:2603.00702, 2026. 7
- [20] Chang Liu, Chun Yang, Hai-Bo Qin, Xiaobin Zhu, Cheng-Lin Liu, and Xu-Cheng Yin. Towards open-set text recognition via label-to-prototype learning. arXiv preprint arXiv:2203.05179, 2022. 3
- [21] Chang Liu, Chun Yang, and Xu-Cheng Yin. Open-set text recognition via character-context decoupling. arXiv preprint arXiv:2204.05535, 2022. 3
- [22] Chang Liu, Simon Corbillé, and Elisa H Barney Smith. Moose: Multi-orientation sharing experts for open-set scene text recognition. arXiv preprint arXiv:2407.18616, 2024. 3
- [23] Harsh Lunia, Ajoy Mondal, and C V Jawahar. Indicstr12: A dataset for indic scene text recognition. arXiv preprint arXiv:2403.08007, 2024. 1, 2, 4
- [24] Mir Tafseer Nayeem, Sawsan Alqahtani, Md Tahmid Rahman Laskar, Tasnim Mohiuddin, and M Saiful Bari. Beyond fertility: Analyzing strr as a metric for

- 784 multilingual tokenization evaluation. arXiv preprint
785 arXiv:2510.09947, 2025. 3, 7
- 786 [25] Vannkinh Nom, Souhail Bakkali, Muhammad Muz-
787 zamil Luqman, Mickaël Coustaty, and Jean-Marc
788 Ogier. Khmerst: A low-resource khmer scene text
789 detection and recognition benchmark. arXiv preprint
790 arXiv:2410.18277, 2024. 2, 7
- 791 [26] Jirui Qi, Raquel Fernández, and Arianna Bisazza.
792 Cross-lingual consistency of factual knowledge in
793 multilingual language models. arXiv preprint
794 arXiv:2310.10378, 2023. 3
- 795 [27] Miao Rang, Zhenni Bi, Chuanjian Liu, Yunhe Wang,
796 and Kai Han. An empirical study of scaling law for
797 ocr. arXiv preprint arXiv:2401.00028, 2023. 3
- 798 [28] Sadab Shipper, Tawsif Tashwar Dipto, Mir Md In-
799 zamam, and Eshat Tanzeem. Banglawild: An in-
800 the-wild bengali scene text recognition benchmark
801 for ocr and vision-language models. arXiv preprint
802 arXiv:2608.03884, 2026. 2
- 803 [29] Bin Xiao, Haiping Wu, Weijian Xu, Xiyang Dai,
804 Houdong Hu, Yumao Lu, Michael Zeng, Ce Liu, and
805 Lu Yuan. Florence-2: Advancing a unified represen-
806 tation for a variety of vision tasks. arXiv preprint
807 arXiv:2311.06242, 2023. 2, 4
- 808 [30] Xingsong Ye, Yongkun Du, JiaXin Zhang, Chen Li,
809 Jing Lyu, and Zhineng Chen. What is wrong with
810 synthetic data for scene text recognition? a strong
811 synthetic engine with diverse simulations and self-
812 evolution. arXiv preprint arXiv:2602.06450, 2026. 2
- 813 [31] Jinshan Zeng, Ruiying Xu, Yu Wu, Hongwei Li, and
814 Jiaying Lu. Star: Zero-shot chinese character recog-
815 nition with stroke- and radical-level decompositions.
816 arXiv preprint arXiv:2210.08490, 2022. 3
- 817 [32] Yuyi Zhang, Yuanzhi Zhu, Dezhi Peng, Peirong Zhang,
818 Zhenhua Yang, Zhibo Yang, Cong Yao, and Lianwen
819 Jin. Hiercode: A lightweight hierarchical codebook
820 for zero-shot chinese text recognition. arXiv preprint
821 arXiv:2403.13761, 2024. 3